RITURAJ SINGH

AI Product Engineer · Full-Stack SaaS Builder · Ex-Tata Steel

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SUMMARY

3rd-year CSE (AI & ML) student at VIT Bhopal with a track record of shipping full-stack AI products end-to-end. Solo-built FileMyCa, a production B2B SaaS for Indian CA firms, in 5 days. Developed computer vision safety systems at Tata Steel; engineered Python automation tooling with 80%+ test coverage. Open to SWE, AI product engineering, and full-stack internships.

TECHNICAL SKILLS

Languages: Python · TypeScript · JavaScript · Java · C++ **AI & ML:** PyTorch · TensorFlow · Scikit-learn · OpenCV · Pandas · Time-series Analysis

Full-Stack: Next.js 15 · React · FastAPI · Supabase · PostgreSQL · MongoDB · SheetJS · Recharts **DevOps:** Docker · AWS · Vercel · Git · GitHub Actions

PROJECTS

FileMyCa — GST Data Collection SaaS

Mar 2026 – Present

Next.js 15 · TypeScript · Supabase · SheetJS · Recharts · Vercel

- Solo-built a production-ready B2B SaaS portal replacing WhatsApp-based GST data collection for Indian CA firms — shipped in 5 days from scaffold to Vercel deploy
- CA admin dashboard with live status tracking across 47 MSME clients, submission charts, deadline countdown, filterable/sortable client table, and slide-out detail panels with status controls
- MSME client upload portal with SheetJS Excel validation (6 required GST column checks), drag-drop Supabase Storage upload, and 4-state upload machine with success animation
- Reminder system with WhatsApp/email template preview modal, bulk send, per-client 60s anti-spam cooldown, reminder history log, and API stubs architected for Gupshup integration
- Custom dark premium CSS design system built from scratch — near-black background, indigo accent, reusable tokens, status badges, card and table component classes; zero Tailwind

eSim Tool Manager — Cross-Platform EDA Automation CLI

Jan – Feb 2026

Python · Subprocess Management · Factory & Singleton Patterns · Unit Testing

- Engineered cross-platform dependency manager automating KiCad, Ngspice and GHDL installation across Windows and Linux — cutting EDA lab setup from 4+ hours to under 10 minutes
- 80%+ test coverage across 1,500 lines of Python with 27 unit tests using safe mocking and subprocess isolation; added macOS support via single adapter class without touching core modules
- Built production-grade CLI with progress indicators, exponential backoff, and status feedback; estimated ₹80,000/semester in engineering hours saved for university EDA labs

WORK EXPERIENCE

AI Intern · Tata Steel Utilities & Infrastructure Services Ltd

May – Jun 2025

Jamshedpur, India

- Built OpenCV + Python computer vision system detecting speeding vehicles from 4 simultaneous CCTV feeds with real-time frame-by-frame analysis and automated alert generation
- Deployed solution processing 1,000+ vehicle detections per day at 92% accuracy, reducing manual safety monitoring workload by ~60% across an active industrial facility

EDUCATION

B.Tech — Computer Science & Engineering (AI & ML) · VIT Bhopal University

Expected May 2027

CGPA: 8.42 / 10.0 · Coursework: Machine Learning · Deep Learning · Computer Vision · NLP · Cloud Computing